Observation Anomaly Detection for Block Adjustment of Satellite Images Based on the Uncertainty Propagation

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***Abstract*—Because of the discrepancies of geometric and radiometric factors, automatic image matching often fails to generate tie points, leading to positioning inconsistencies in satellite image blocks. To address this issue, we proposed a novel observation anomaly detection method based on uncertainty propagation and clustering analysis. The method propagates the geometric precision of a reference image throughout the image block under the intersection and elevation constraints. A feature space is then constructed based on the statistics of the geometric accuracies, and the density-based spatial clustering of applications with noise (DBSCAN) clustering algorithm is applied to distinguish anomalous images from normal ones. Experiments using 149 multispectral and 298 panchromatic GF-7 images demonstrated that the method achieved 100% detection accuracy under the optimal image-space compensation model and the optimal feature space. While some moderate observation anomalies were missed with a single reference image, incorporating multiple well-distributed reference images enabled accurate detection. Further analysis revealed significant discrepancies in positioning accuracies between anomalous and normal images after block adjustment, with mean differences up to 4.77 and 5.40 m in the line and sample directions, respectively. By adding approximately 1300 tie points to anomalous images, spatial connectivity was restored, aligning their precision with that of normal images.**

***Index Terms*—Block adjustment, density-based spatial clustering of applications with noise (DBSCAN), observation anomaly, tie points, uncertainty propagation law.**

I. Introduction

SATELLITE imagery provides large-scale spatial information essential for advanced applications such as global mapping [1], 3-D modeling [2], change detection [3], and disaster monitoring. It is crucial to ensure the accuracy of this spatial information, so after a satellite is launched, onboard calibration is conducted to correct systematic image biases [4]. However, random errors caused by factors such as attribute jitter, thermal deformation, and other unpredictable influences remain uncorrected. These errors lead to inconsistencies in positioning accuracy within the image block. Block adjustment addresses this issue: this process involves using the intersections of homologous rays to generate an image block with high positioning accuracy. Thus, block adjustment is a vital component in the high-accuracy processing of satellite imagery.

In the past decade, advancements in satellite sensors have addressed the increasing demand for high-accuracy spatial information essential for global sustainable development. Block adjustment techniques have evolved from using ground control points (GCPs) to GCP-free methods, in which image block factors, such as observation conditions, image overlap, and prior accuracy, become more influential [5]. Currently, automatic image matching is employed to observe tie points within image blocks [6]. Gross errors are effectively eliminated when sufficient observations are included, but the differences in imaging time, observation angle, and radiometric quality lead to the uneven distribution or lack of tie points in certain areas, resulting in observation anomalies. The differences cannot be detected through tie point statistics alone.

Previous analyses have concluded that three main types of observation anomaly may be present in an image block.

1) The first is the weak connection of a single image, in which tie points are distributed extremely unevenly within the image. In the example shown in Fig. 1(a), tie points are concentrated only in the top right corner of the image, possibly due to land use differences.
2) The second type of observation anomaly is the weak connection of multiple images. In the example shown in Fig. 1(b), two well-connected image blocks have only sparse tie points within the overlapping area; this phenomenon occurs when images within the block are captured at different times, with ground truth changes between image captures.
3) The third type is the disconnection of multiple images. The example shown in Fig. 1(c) shows two well-connected image blocks with no tie points within the overlapping area, which can be caused by a weak ground texture, such as forest or desert.

The presence of observation anomalies does not affect the block adjustment process, but they can trap the positioning accuracy of anomalous images in local minima, leading to results that deviate from the remainder of the block. Such

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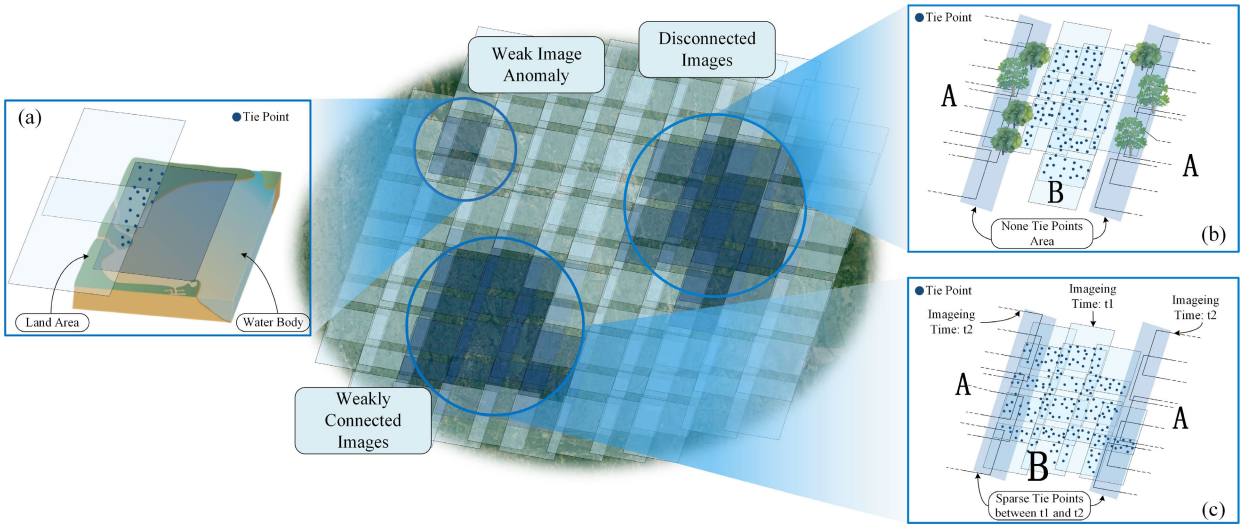


Fig. 1. Schematic of observation anomalies in the image block. (a) Single-weakly-connected-image observation anomaly (the dark blue rectangle) and normal images (light blue rectangles). (b) Multiple-weakly-connected-images observation anomaly (blue rectangles) and normal images (lines). (c) Multiple-disconnected-images (blue rectangles) and normal images (lines).

discrepancies lead to inconsistent accuracy of the block adjustment results, presenting a significant quality issue, particularly for large-scale image blocks. Because of the complex topographic relationships within the image block, it is difficult to locate observation anomalies. Commonly used quality control methods identify quality issues through statistics of tie point positioning accuracy before and after block adjustment, such as computing back-projection error. Nevertheless, insufficient assessment of the number, distribution, or positioning accuracy of tie points prior to block adjustment prevents the effective identification of observation anomalies in the image block for two reasons. First, observation anomalies are related to observation conditions and the mathematical model used for block adjustment. For example, experiments have revealed that when a translation model is used to compensate for a weakly connected image, accurate parameters can be obtained when a sufficient number of tie points is used, even if they are unevenly distributed. However, affine or higher-order models under the same conditions yield imprecise or error-prone parameters. Second, an observation anomaly cannot simply be equated to the sparsity or lack of tie points on an image, and therefore cannot be detected by only counting the tie points on each image. Similarly, after block adjustment, observation anomalies within the image block cannot be detected through either simple accuracy statistics, such as the median error of estimated parameters or tie point residuals, or manual validation.

In any adjustment system, the variance of observations directly reflects the observation conditions [7]. Therefore, we developed a novel method based on the rational function model (RFM) for detecting observation anomalies based on the propagation of uncertainty in the accuracy of the image-space compensation model, combined with clustering analysis. This method sets a reference image to establish an initial geometric accuracy for the image block and propagates this accuracy to other images using intersection and elevation constraints. Next, the density-based spatial clustering of applications with noise (DBSCAN) algorithm is employed to identify images exhibiting observation anomalies. This approach enables the identification of all anomalous images prior to block adjustment. Using traditional quality assessment methods, the compensation model parameter accuracy can be computed from the inverse of the coefficient matrix of the normal equation, but it cannot be applied directly to the proposed detection method, unless the following two issues are solved. First, the differentiation of the compensation model parameter accuracy of the anomalous image and the normal image must be achieved. Block adjustment introduces extra observations, particularly when GCPs are not absent. For example, Yang et al. [8] added virtual control points as constraints to ensure the robustness of a block adjustment system under GCP-free conditions. However, if the image is constrained by virtual control points, both the anomalous and normal images will have uniformly high compensation model parameter accuracy that will overwhelm the differentiation. In block adjustment, the planimetric and elevation accuracies of images are usually coupled [9]. The theoretical accuracy of the reference image propagates in both directions, such that even a normal image may have a theoretical accuracy as low as that of an anomalous image. The second issue involves determining the geometric accuracy of one image. In a block adjustment system, geometric accuracy can be represented by the compensation model parameter variance, but it is strongly correlated with the model's degree and lacks universality.

The remainder of this article is arranged as follows. Section II presents a review of common quality control methods, their features, and introduces the core concepts of the RFM model, as well as the uncertainty propagation law. Section III describes the methodology of detecting observation anomalies, including the establishment of the intersection and elevation constraints, the geometric accuracy of an image described by

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the image-space compensation model, and the determination of boundary conditions by the DBSCAN algorithm. Section IV describes the data and study site in detail, together with the experiments conducted to validate our observation anomaly detection model and the negative effects of observation anomalies on block adjustment. Section V involves the summary and conclusion.

In summary, we developed an observation anomaly detection method that leverages the uncertainty propagation law and the DBSCAN clustering algorithm. The contributions of our method can be mainly concluded as follows.

1) We proposed an effective observation anomaly detection method based on the discussions and analysis of the observation anomalies, by analyzing the source and the features of the observation anomaly in the image block. Through constructing the intersection constraint and the elevation constraint, ensuring the initial geometric accuracy propagates in the image block primarily along the planimetric direction, and can be consequently used to distinguish the anomalous images effectively. In order to establish the boundary condition of normal images and anomalous images precisely, we applied the DBSCAN algorithm in the mean-max feature space. Our method commendably achieves the detection of observation anomalies in the planimetric and stereo image blocks.
2) We analyzed the impact of the observation anomalies on the uncontrolled block adjustment and formulated a feasible method to improve the positioning accuracy. All kinds of observation anomalies are added to the planimetric and stereo image blocks, and they degrade the positioning accuracy at different levels. To optimize the internal positioning accuracy consistency of the image block, we add tie points to the anomalous regions, and their positioning accuracy significantly improves, providing an effective approach to optimizing the block adjustment result after detecting the observation anomalies.
3) We proposed a novel method to enhance the quality-control framework for the satellite image block adjustment. Not relying on the block adjustment, our method achieves the effective and accurate detection of observation anomalies before the adjustment process, and is completely performed without GCPs, providing a time-saving and cost-effective block adjustment quality control method. It fills the current gap in quality control methods for automatic and accurate detection of observational anomalies.

II. Related Works

Bundle block adjustment can process numerous satellite images, sometimes including multi-modal images, yielding and updating more than 70%–80% of 1:50 000 maps maintained worldwide, and providing fundamental data for advanced applications [10]. Thousands of satellite images have extensive spatial coverage, but also quality issues. Although current quality control methods are sufficient for common issues such as gross error elimination [11] and reliability analysis [13], a method specifically designed to detect observation anomalies is needed. The method is introduced systematically in Section I—it is caused by the distribution and the amount of tie points and leads to the positioning accuracy inconsistency of the image block by influencing the structure of the block adjustment model [4], thereby the misalignment exists within mapping products. The RFM is typically used for satellite image block adjustment due to its high-precision fitting of the rigorous imaging model with equivalent positioning accuracy, as well as its accessibility and universality [14], [15]. It establishes the relationship between the object space and the image space [16]. Additionally, image-space compensation models, are usually applied to compensate for the offset between images at the position (l, s), together with image distortion during block adjustment; their forms are expressed as shown in the following equation:

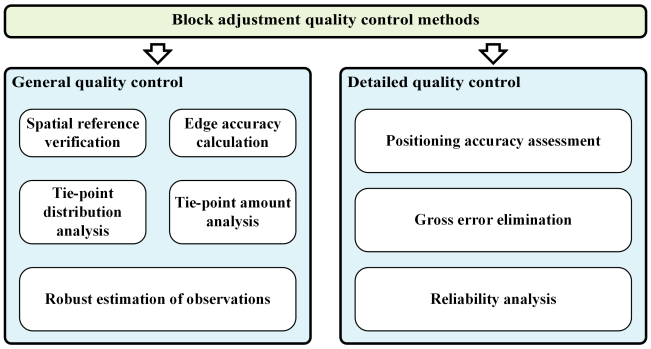


Fig. 2. Categories of block adjustment quality control methods.

$$\begin{cases} \Delta l = \sum_{i=0}^{n} \sum_{j=0}^{n-i} a_{ij} l^i s^j \\ \Delta s = \sum_{i=0}^{n} \sum_{j=0}^{n-i} b_{ij} l^i s^j \end{cases} \tag{1}$$

where Δl and Δs are image-space compensation models a_{ij} and b_{ij} are the coefficients of the image compensation model and the degree i and j can be adjusted.

The RFM is the most general block adjustment model, and is commonly used to produce high-accuracy geo-information products [5], [17]; it is the fundamental model of our study.

Existing block adjustment quality control methods can be divided into two categories: general quality control and detailed quality control (Fig. 2) [18]. The former category includes both qualitative and quantitative methods, such as reference system verification (qualitative) and edge accuracy assessment (quantitative). General quality control approaches are usually GCP-free; for example, edge accuracy assessment calculates relative accuracy after block adjustment by manually checking the alignment of overlapped images or computing the back-propagation accuracy of tie points, such that there is no need for ground information provided by GCPs. In contrast, detailed quality control methods are more quantitative. In these approaches, GCPs or other reference data of equal or higher accuracy [19] are required to assess the absolute positioning accuracy of the block adjustment [8]. One commonly used approach is to select evenly distributed GCPs as check-points and calculate their back-projection error to

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estimate the planimetric or elevation accuracy of the block adjustment. Multi-modal satellite data are sometimes applied to assess positioning accuracy in detailed quality control [20], but they are more often used to improve block adjustment positioning accuracy. However, neither of these two quality control methods is a feasible approach for detecting observation anomalies. Estimation of the relative positioning accuracy of block adjustment based on tie point statistics (general quality control methods) can only detect gross tie point errors, which are easily eliminated by ranndom sample consensus (RANSAC) algorithm or other robust estimation approaches [21], [22]. Even estimation approaches based on the number and distribution of tie points cannot locate all observation anomalies. In contrast, detailed quality control methods can theoretically locate the anomalous image if given an ideal distribution of GCPs and checkpoints; however, such a perfect configuration is almost impossible in practice, particularly when GCPs are absent, where spatial connection is provided by tie points alone, and the observation anomaly cannot be reflected by the back-projection error of tie points. Therefore, a method must be designed specifically to detect observation anomalies prior to block adjustment.

The uncertainty propagation law is applicable to this problem. It describes the propagation of errors, being widely used in image processing and machine learning [26]. In surveying and mapping, the law is applied to assess the quality of surveying control networks [25], global navigation satellite system data [26], and LiDAR data processing [27]. It is also used in photogrammetry, particularly in stereo vision and 3-D modeling. For example, Ren et al. [28] developed a method for analyzing uncertainty propagation in stereo triangulation by designing an efficient factorial design. Based on the results of their factorial experiments, they estimated the covariance matrix of the 3-D coordinates of measurement points, from which measurement uncertainties were derived along the x, y, and z-axes, as well as in the principal axis direction. Comparison of these results with those obtained using the Monte Carlo method revealed that this approach achieved comparable precision while significantly reducing computational costs. Similarly, Polic and Pajdla [29] introduced a novel method to enhance the numerical precision of uncertainty propagation in large-scale structure from motion tasks. This approach efficiently managed datasets comprising millions of feature points recorded by thousands of cameras, resulting in millions of 3-D points stored on a single computer. The method was validated through comparative experiments using ground truth data. Hagemann et al. [30] presented a unified framework for evaluating camera calibration errors by separating systematic bias from random uncertainty, in which a resampling estimator quantified uncertainty beyond classical covariance, and a model-agnostic metric enabled robust evaluation. The framework was validated using both simulations and real cameras, and was found to support both individual calibration checks and benchmarking. However, all of these methods primarily applied the uncertainty propagation law to quality assessment, leaving further exploration of its potential in anomalous data detection largely unaddressed. As a specialized approach in satellite photogrammetry, few studies have used the uncertainty propagation law to analyze the propagation of accuracy in image blocks.

Therefore, we innovatively used the uncertainty propagation law to detect images with observation anomalies in the image block after analyzing the characteristics of satellite imaging. We first presented the geometric accuracy of the images by the accuracy of the image-space compensation model parameters. Next, we selected the reference image(s) and added virtual observations to provide an initial geometric accuracy for the entire block. Additionally, we constructed intersection and elevation constraints for the RFM with image-space compensations, ensuring that the initial geometric accuracy propagates primarily along the planimetric direction. Third, we created grids on each image and computed the accuracy of the compensation model at grid points to precisely describe the geometric accuracy within an image. The overall accuracy of the entire image can consequently be obtained by aggregating the statistics of the geometric accuracy at grid points. Finally, we established the feature space based on the calculated statistics and performed DBSCAN to accurately identify the anomalous images.

III. Methodology

A. Overview

Our study is the first to use the uncertainty propagation law to detect observation anomalies within an image block, after analyzing satellite imagery characteristics. First, we designated a reference image and enhanced its geometric accuracy by virtual observations. After analyzing the uncertainty propagation characteristics of the image block, we established the elevation constraint to address the challenge posed by the coupling of planimetric and elevation accuracies, which hinders accuracy propagation. Next, we developed an observation anomaly detection model and calculated the parameter accuracy of the compensation model for each image. According to the uncertainty propagation law, we subsequently computed the accuracy of the compensation model to quantitatively represent the observation conditions both within an individual image and across the entire image block. Finally, we derived statistics of the compensation model accuracy for each image; some we are used as features to construct the feature space. The DBSCAN algorithm was then applied to classify anomalous and normal images. Fig. 3 presents the pipeline for our entire methodology.

B. Establishment of Reference Image Geometric Accuracy Model

In an adjustment system, the observation conditions are fundamentally represented by the accuracy of block adjustment parameters. However, with only this metric, anomalous and normal images cannot be precisely differentiated. To address this issue, we designate a reference image (e.g., the image at the center of the image block) or multiple reference images (distributed evenly). A central reference image allows the theoretical accuracy propagation to the whole image block in the radial direction. The theoretical accuracies of different levels form concentric circles with the reference image as the

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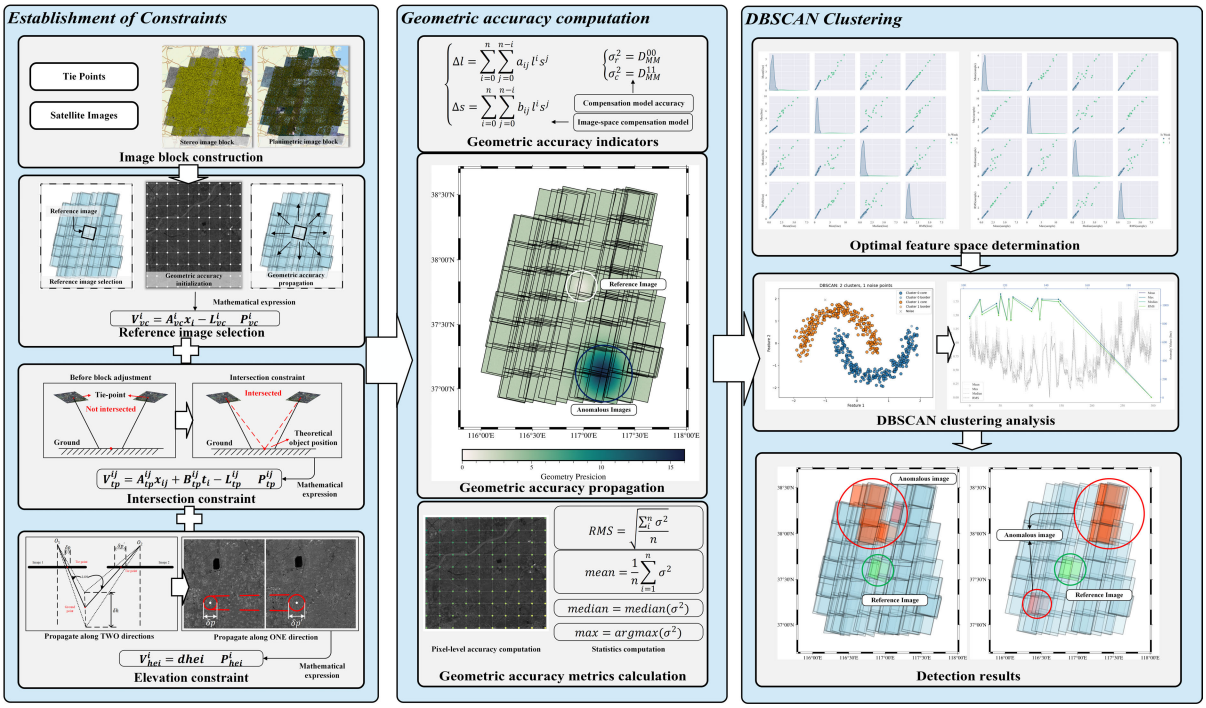


Fig. 3. Pipeline of our method.

center. Therefore, the theoretical accuracy on the border area of the image block may not be extremely low. When the size of the image block is large, the theoretical accuracy of the central image could also degrade severely and cause misclassification. It is feasible to select evenly distributed images as references to solve this problem. Subsequently, an image grid is created to establish virtual observations for the compensation model, which is described as shown in the following equation:

$$\begin{cases} G_y^k = \Delta l\,(l,s) \\ G_x^k = \Delta s\,(l,s) \end{cases} \tag{2}$$

where $\Delta l(\cdot)$ and $\Delta s(\cdot)$ represent the compensation models for the image position at coordinates (l, s), G_y^k and G_x^k are the corresponding virtual observation adjustment values.

Because of virtual observations, the accuracy of the reference image is relatively high. This geometric accuracy propagates to other images within the image block. Images that are connected, directly or indirectly, to the reference image have high accuracy, while those that are weakly connected or disconnected from the reference image have relatively low geometric accuracy. Fig. 4 illustrates this phenomenon.

After linearization, the error function of the virtual observation at the ith grid point on the reference image can be expressed as shown in the following equation:

$$\begin{aligned} \boldsymbol{V_{vc}^i} &= \boldsymbol{A_{vc}^i x_i} - \boldsymbol{L_{vc}^i} \cdots \boldsymbol{P_{vc}^i} \\ \boldsymbol{A_{vc}^i} &= \begin{bmatrix} \dfrac{\partial G_y^i}{\partial a_0} & \cdots & \dfrac{\partial G_y^i}{\partial a_n} & \dfrac{\partial G_y^i}{\partial b_0} & \cdots & \dfrac{\partial G_y^i}{\partial b_n} \\ \dfrac{\partial G_x^i}{\partial a_0} & \cdots & \dfrac{\partial G_x^i}{\partial a_n} & \dfrac{\partial G_x^i}{\partial b_0} & \cdots & \dfrac{\partial G_x^i}{\partial b_n} \end{bmatrix} \\ \boldsymbol{L_{tp}^{ij}} &= \begin{bmatrix} -G_y^i \\ -G_x^i \end{bmatrix} \end{aligned} \tag{3}$$

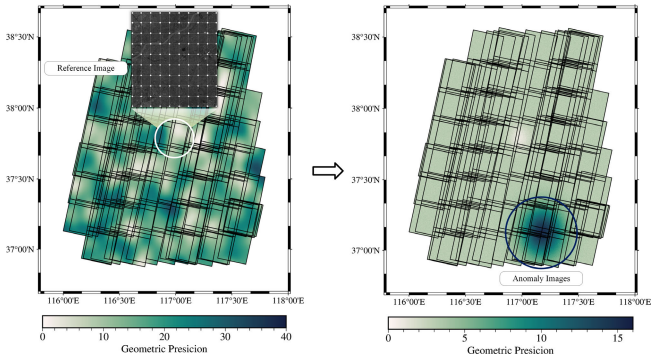


Fig. 4. Schematic of the theoretical-accuracy-control effect of the reference image (unit: pixel).

where $\boldsymbol{x_i}$ is the compensation model parameter vector of the ith grid point, $\boldsymbol{A_{vc}^i}$ is the coefficient matrix, $\boldsymbol{L_{vc}^i}$ is the constant vector, and $\boldsymbol{P_{vc}^i}$ is the weight matrix. Typically, the number of grid points is lower than that of tie points in a single image. To ensure high geometric accuracy of the reference image, we set large weights to virtual observations, approximately 10 000 times greater than those of tie points. This ensures balancing the influence between the two kinds of observations.

C. Establishment of the Intersection and Elevation Constraints

We further developed constraints that ensure the propagation of geometric accuracy along the planimetric direction. The first is the intersection constraint. Although homologous rays do not intersect prior to block adjustment, tie points ideally correspond to the same object coordinates. To accurately describe the constraint, we designate the jth image point of

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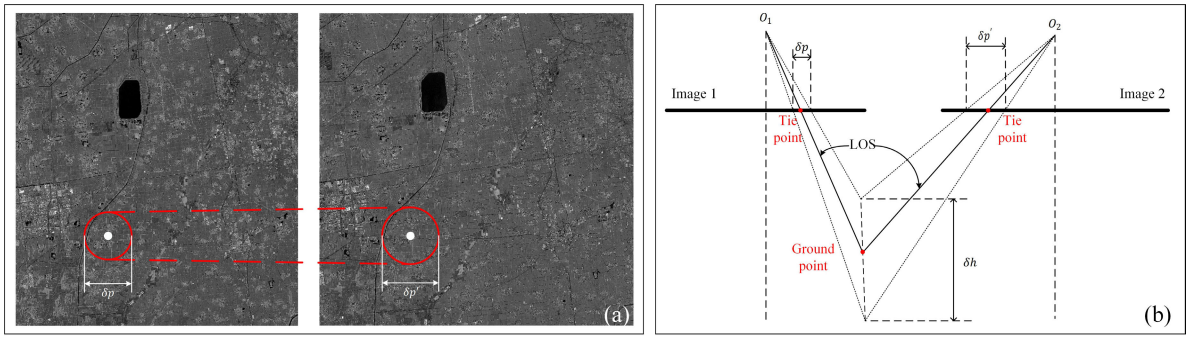


Fig. 5. Propagation of initial geometric accuracy in both ideal and practical situations. (a) Ideally, the accuracy propagates solely in the planimetric direction. (b) Practically, the accuracy propagates in both the planimetric and vertical directions.

the ith tie point as tp_{ij}. The intersection constraint is, as shown in the following equation:

$$\begin{cases} G_y^{ij} = F_y(Lat, Lon, Hei) - l - \Delta l \\ G_x^{ij} = F_x(Lat, Lon, Hei) - s - \Delta s \end{cases} \tag{4}$$

Through linearization, the corresponding error equation is expressed, as shown in the following equation:

$$\begin{aligned} \boldsymbol{V}_{tp}^{ij} &= \boldsymbol{A}_{tp}^{ij}\boldsymbol{x}_{ij} + \boldsymbol{B}_{tp}^{ij}\boldsymbol{t}_i - \boldsymbol{L}_{tp}^{ij} \cdots \boldsymbol{P}_{tp}^{ij} \\ \boldsymbol{A}_{tp}^{ij} &= \begin{bmatrix} \dfrac{\partial G_y^{ij}}{\partial a_0} & \cdots & \dfrac{\partial G_y^{ij}}{\partial a_n} & \dfrac{\partial G_y^{ij}}{\partial b_0} & \cdots & \dfrac{\partial G_y^{ij}}{\partial b_n} \\ \dfrac{\partial G_x^{ij}}{\partial a_0} & \cdots & \dfrac{\partial G_x^{ij}}{\partial a_n} & \dfrac{\partial G_x^{ij}}{\partial b_0} & \cdots & \dfrac{\partial G_x^{ij}}{\partial b_n} \end{bmatrix} \\ \boldsymbol{B}_{tp}^{ij} &= \begin{bmatrix} \dfrac{\partial G_y^{ij}}{\partial Lat} & \dfrac{\partial G_y^{ij}}{\partial Lon} & \dfrac{\partial G_y^{ij}}{\partial Hei} \\ \dfrac{\partial G_x^{ij}}{\partial Lat} & \dfrac{\partial G_x^{ij}}{\partial Lon} & \dfrac{\partial G_x^{ij}}{\partial Hei} \end{bmatrix}, \quad \boldsymbol{L}_{tp}^{ij} = \begin{bmatrix} -G_y^{ij} \\ -G_x^{ij} \end{bmatrix} \end{aligned} \tag{5}$$

where $\boldsymbol{x_{ij}}$ is the compensation model parameter adjustment vector of the image where tp_{ij} is located, $\boldsymbol{t_i}$ represents the object coordinates of the ith tie point, $\boldsymbol{A}_{tp}^{ij}$ is the coefficient matrix of the compensation model parameter, $\boldsymbol{B}_{tp}^{ij}$ is the coefficient matrix of tie point ground coordinates, $\boldsymbol{P}_{tp}^{ij}$ is the weight matrix of observations, and n is the number of compensation model parameters.

Considering the tie points of all images in the image block, the constraint for the ith tie point is established, as shown in the following equation:

$$\boldsymbol{V}_{tp}^{i} = \boldsymbol{A}_{tp}^{i}\boldsymbol{x}_i + \boldsymbol{B}_{tp}^{i}\boldsymbol{t}_i - \boldsymbol{L}_{tp}^{i} \cdots \boldsymbol{P}_{tp}^{i} \tag{6}$$

where $\boldsymbol{V}_{tp}^{i}$ is the residual vector of the observation vector, $\boldsymbol{A}_{tp}^{i}$ is the coefficient matrix of adjustment parameters, $\boldsymbol{B}_{tp}^{i}$ is the coefficient matrix of object coordinates, $\boldsymbol{L}_{tp}^{i}$ is the constant vector of the error function, and $\boldsymbol{P}_{tp}^{ij}$ is the weight matrix.

The second constraint is the elevation constraint. Previous studies have demonstrated coupling between the planimetric and elevation accuracies of intersected images, which indicates that the high geometric accuracy in the planimetric direction (i.e., the accuracy of the compensation model parameters) of the reference image may propagate through the image block in both the planimetric and vertical directions. This coupling may even result in a significant decline in planimetric accuracy (Fig. 5), which can lead to a normal image being misclassified as anomalous. We aim for the initial geometric accuracy provided by the reference image to propagate solely along the planimetric direction within the image block, to ensure that it is reflected in the image-space compensation model to the greatest extent.

In the ideal scenario [Fig. 5(a)], although the geometric accuracy δp and $\delta p'$ between two images with firm spatial connections may be affected by geometric factors (e.g., imaging angle), the difference in accuracy is generally minimal. However, in practice, the planimetric geometry is influenced by the elevation accuracy [31], and the former probably degrades when propagating from one image to another. As shown in Fig. 5(b), the geometric accuracy $\delta p'$ of image 2, which is propagated from image 1, degrades considerably due to the influence of the elevation uncertainty δh. To address this issue, we perform virtual observation on the tie point elevation and introduce a pre-set elevation accuracy. Constrained by this, the geometric accuracy will primarily propagate in the planimetric direction. The elevation constraint of the ith tie point is expressed, as shown in the following equation:

$$\boldsymbol{V}_{hei}^{i} = \boldsymbol{dhei} \cdots \boldsymbol{P}_{hei}^{i} \tag{7}$$

where $\boldsymbol{V}_{hei}^{i}$ is the residual vector of the elevation constraint, $\boldsymbol{dhei}$ is the adjustment vector of the tie point elevation, and $\boldsymbol{P}_{hei}^{i}$ is the weight matrix.

By integrating the initial geometric accuracy of the reference image, the intersection constraint, and the elevation constraint, the geometric accuracy estimation model for each image is derived, as shown in the following equation:

$$\begin{cases} \boldsymbol{V}_{tp}^{i} = \boldsymbol{A}_{tp}^{i}\boldsymbol{x}_i + \boldsymbol{B}_{tp}^{i}\boldsymbol{t}_i - \boldsymbol{L}_{tp}^{i}, & \boldsymbol{P}_{tp}^{i} \\ \boldsymbol{V}_{hei}^{i} = \boldsymbol{dhei}, & \boldsymbol{P}_{hei}^{i} \\ \boldsymbol{V}_{vc}^{i} = \boldsymbol{A}_{vc}^{i}\boldsymbol{x}_i - \boldsymbol{L}_{vc}^{i}, & \boldsymbol{P}_{vc}^{i} \end{cases} \tag{8}$$

where $\boldsymbol{x}$ is the adjustment vector of the block adjustment parameters, $\boldsymbol{t}$ is the adjustment vector of the tie point object coordinates, $\boldsymbol{V}_{tp}$ and $\boldsymbol{V}_{vc}$ are the residual vectors of the intersection constraint and the virtual observation equation of the reference image, respectively, $\boldsymbol{A}_{tp}$ and $\boldsymbol{A}_{vc}$ are the coefficient matrices corresponding to the equations with respect to the block adjustment parameters, $\boldsymbol{B}_{tp}$ is the coefficient matrix of the intersection constraint with respect to the object

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coordinates of the tie points, $\boldsymbol{L_{tp}}$ and $\boldsymbol{L_{vc}}$ are constant vectors, and $\boldsymbol{P_{tp}}$ and $\boldsymbol{P_{vc}}$ are the corresponding weight matrices.

D. Geometric Accuracy Described by the Image-Space Compensation Model

Based on the model presented in Section III-C, the accuracy of the image-space compensation model parameters can be used to describe the geometric accuracy of the image. For a block adjustment system, the error function is

$$\begin{aligned}
&\boldsymbol{V} = \boldsymbol{Ax} + \boldsymbol{Bt} - \boldsymbol{L} \cdots \boldsymbol{P} \\
&\boldsymbol{V} = \begin{bmatrix} V_p \\ V_r \\ V_g \end{bmatrix}, \quad \boldsymbol{A} = \begin{bmatrix} A_{tp} \\ A_{vc} \\ 0 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} B_{tp} \\ 0 \\ B_g \end{bmatrix} \\
&\boldsymbol{L} = \begin{bmatrix} L_{tp} \\ L_{vc} \\ 0 \end{bmatrix} \\
&\boldsymbol{P} = \begin{bmatrix} P_{tp} & O & O \\ O & P_{vc} & O \\ O & O & P_g \end{bmatrix}, \quad \boldsymbol{B_g} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&\boldsymbol{P_g} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \dfrac{1}{\sigma_{hei}^2} \end{bmatrix}
\end{aligned} \tag{9}$$

where $\boldsymbol{V}$ is the adjustment vector of the observations, $\boldsymbol{x}$ and $\boldsymbol{t}$ are adjustment vectors for the block adjustment parameter and the object-space coordinates of the tie points, $\boldsymbol{x}$ is usually the compensation model parameter adjustment vector, $\boldsymbol{L}$ is the constant vector, and $\boldsymbol{P}$ is the weight matrix.

According to the error adjustment theory, the accuracy of arbitrary adjustment parameters can be computed from the coefficient matrix of the normal equation. The normal equation of (9) can be expressed, as shown in the following equation:

$$\begin{bmatrix} \boldsymbol{A^T PA} & \boldsymbol{A^T PB} \\ \boldsymbol{B^T PA} & \boldsymbol{B^T PB} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{t} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A^T PL} \\ \boldsymbol{B^T PL} \end{bmatrix} \tag{10}$$

Applying Gaussian elimination, the coefficient matrix of the normal equation with regard to the block adjustment parameter $\boldsymbol{x}$ can be calculated by eliminating the ground coordinate adjustment vector $\boldsymbol{t}$ of the tie point in (10), and its inverse matrix $\boldsymbol{Q_{xx}}$ is the cofactor matrix for $\boldsymbol{x}$, as shown in the following equation:

$$\begin{cases} \boldsymbol{Q_{xx}} = \left(\boldsymbol{N_{AA}} - \boldsymbol{N_{AB}}(\boldsymbol{N_{BB}})^{-1}\boldsymbol{N_{BA}}\right)^{-1} \\ \boldsymbol{N_{AA}} = \boldsymbol{A^T PA} = \boldsymbol{A_{tp}^T P_{tp} A_{tp}} + \boldsymbol{A_{vc}^T P_{vc} A_{vc}} \\ \boldsymbol{N_{AB}} = \boldsymbol{A^T PB} = \boldsymbol{A_{tp}^T P_{tp} B_{tp}} \\ \boldsymbol{N_{BB}} = \boldsymbol{B^T PB} = \boldsymbol{B_{tp}^T P_{tp} B_{tp}} + \boldsymbol{B_g^T P_g B_g} \\ \boldsymbol{N_{BA}} = \boldsymbol{B^T PA} = \boldsymbol{B_{tp}^T P_{tp} A_{tp}}. \end{cases} \tag{11}$$

Then the covariance matrix $\boldsymbol{D_{xx}}$ is calculated, as shown in the following equation:

$$\boldsymbol{D_{xx}} = \sigma_0^2 \boldsymbol{Q_{xx}} \tag{12}$$

where σ_0 is the standard error of unit weight. Because this value typically corresponds to the tie point, we set it as a constant.

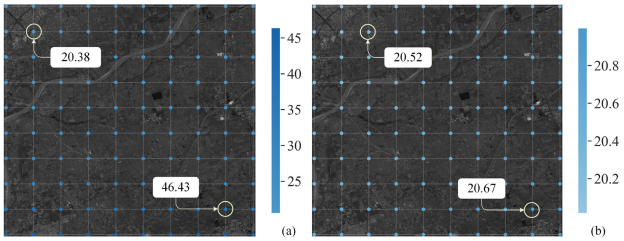


Fig. 6. Comparison of the geometric accuracies of anomalous and normal images (unit: pixel). (a) Anomalous image. (b) Normal image.

According to the uncertainty propagation law, the covariance of the block adjustment parameters is identical to that of their adjustments. Consequently, the diagonal elements of the covariance matrix $\boldsymbol{D_{xx}}$ for the adjustment values correspond precisely to the variance σ_i^2 of the block adjustment parameters. This relationship can be used to represent geometric accuracy; it is expressed as shown in the following equation:

$$\sigma_i^2 = \boldsymbol{D_{ii}} = \sigma_0^2 \boldsymbol{Q_{ii}} \tag{13}$$

where $\boldsymbol{Q_{ii}}$ and $\boldsymbol{D_{ii}}$ are the ith diagonal elements of matrices $\boldsymbol{Q_{xx}}$ and $\boldsymbol{D_{xx}}$.

E. Compensation Model Accuracy Estimation Taking Into Account Pixel Position

The block adjustment parameter coefficient matrix computed in Section III-D represents only the overall geometric accuracy of an image. Observation anomalies may also result in significant differences in geometric accuracy within an image, as shown in Fig. 6.

To accurately express differences in geometric accuracy within an image, we create a grid on each image and calculate the accuracy of the compensation model $\Delta r, \Delta c$ at the ith grid point (r, c) according to the uncertainty propagation law. This grid accuracy corresponds to the diagonal elements of the covariance matrix $\boldsymbol{D_{MM}}$, the calculations are, as shown in the following equations:

$$\boldsymbol{D_{MM}} = \boldsymbol{M D_{xx} M^T} \tag{14}$$

$$\begin{aligned}
&\boldsymbol{M} = \begin{bmatrix} \dfrac{\partial \Delta r}{\partial a_0} & \cdots & \dfrac{\partial \Delta r}{\partial a_n} & \dfrac{\partial \Delta c}{\partial a_0} & \cdots & \dfrac{\partial \Delta c}{\partial a_n} \\ \dfrac{\partial \Delta r}{\partial b_0} & \cdots & \dfrac{\partial \Delta r}{\partial b_n} & \dfrac{\partial \Delta c}{\partial b_0} & \cdots & \dfrac{\partial \Delta c}{\partial b_0} \end{bmatrix} \\
&\begin{cases} \sigma_r^2 = D_{MM}^{00} \\ \sigma_c^2 = D_{MM}^{11} \end{cases}
\end{aligned} \tag{15}$$

where D_{MM}^{00} and D_{MM}^{11} are the diagonal elements of the matrix $\boldsymbol{D_{MM}}$.

Our method can accurately represent the distribution of geometric accuracy within an image, enabling the detection of observation anomalies within a single image. It also enables quantification of the overall geometric accuracy of the image by assessing the geometric accuracy of each grid point, which can be used for the subsequent identification of anomalous images. We calculate the mean as $\bar{\sigma^2} = (1/n)\sum_{i=1}^{n} \sigma^2$, the median as $\sigma_{med}^2 = median(\sigma^2)$, the maximum as

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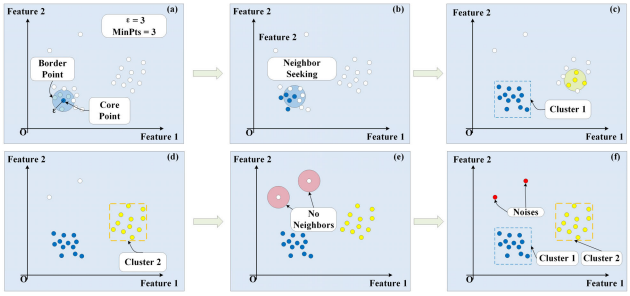


Fig. 7. Schematic of DBSCAN algorithm. (a) Parameter Initialization and Core Point Identification. (b) Neighborhood Expansion from the Core Point. (c) Formation of the First Cluster. (d) Expansion of Another Dense Region. (e) Detection of Isolated Points. (f) Final Clustering Result with Noise Identification.

$\sigma^2_{\max} = \arg\max(\sigma^2)$, and the root-mean-square error (RMSE) as $RMSE_{\sigma^2} = \sqrt{(\sum_i^n(\sigma^2 - \overline{\sigma^2})^2)/n}$, where n is the number of grid points. Anomalous images are detected based on these statistics.

F. Observation Anomaly Boundary Condition Identification by DBSCAN Clustering

In practice, some anomalous images have accuracies that are nearly indistinguishable from those of normal images; it is impossible to detect anomalous images simply by setting a threshold. If we treat observation anomaly detection as a classification task, normal images will cluster together, with anomalous images forming one or more separate groups. There are many mature classification methods [32], but considering the relatively small amount of samples and the simple feature space structure, we apply the DBSCAN algorithm, which is based on sample density characteristics, and combines robust clustering quality with high computational efficiency. Compared with partition-based and hierarchical clustering methods, DBSCAN requires less prior knowledge; it is not similar to K-means, in which the number of clusters is required in advance.

The DBSCAN algorithm is based on the idea that objects in a cluster should be closely connected at high densities, whereas outliers have lower densities. Two core parameters are the neighborhood radius ε, determining the density, and the threshold $MinPts$, which identifies core points. A search template defined by ε and $MinPts$ starts from a given core point and iteratively expands into density-reachable points. This process constructs a maximal area encompassing both core points and border points, in which any two points maintain a density-connected relationship. Clustering of sample points is achieved by iteratively selecting core points and determining whether they are density-reachable.

To detect anomalous images, we compute the mean and maximum values of grid-point geometric accuracy in both the line and sample directions for each image. The mean captures the expected geometric accuracy, and the maximum reflects its dispersion. These two metrics define a 2-D feature space where each image is treated as a sample. We prioritize the mean as the primary feature and use the maximum as the secondary feature to construct the feature space. Subsequently, we perform DBSCAN clustering in the feature space to distinguish normal images from anomalous images. This approach effectively identifies anomalies without requiring predefined cluster counts. Fig. 7 presents a schematic of the DBSCAN algorithm.

TABLE I
DATA USAGE AND TECHNICAL PARAMETERS OF THE GF-7 PLATFORM

	Stereo image block	Planimetric image block
Image number	298	149
Acquisition time	2021.1~2022.12	
Resolution	0.65m (backward)/ 0.8m (forward)	2.6m
Center position	116.93E，37.66N	116.94E，37.69N
Coverage	29,134.33km^2	28,639.95km^2
Elevation range	-5m~40m	

IV. EXPERIMENTS AND ANALYSIS

A. Dataset and Study Area

The image data were acquired by Gaofen-7 (GF-7), China's first civilian sub-meter high-resolution optical stereo mapping satellite. Gaofen-7 is equipped with a dual-line array panchromatic stereo camera, a four-band multispectral camera, a laser altimeter, and other payloads. The dual-line array camera captures panchromatic stereo images with a swath width of 20 km and a spatial resolution better than 0.8 m, while the multispectral images have a resolution of 3.2 m.

To evaluate the effectiveness of our observation anomaly detection method on both planimetric and stereo image blocks, we conducted experiments using Gaofen-7 panchromatic and multispectral images from the same region. The planimetric image block includes 149 multispectral images, and the stereo image block consists of 298 panchromatic images organized into 149 image pairs. The dataset spans acquisition dates from January 2021 to December 2022, covering more than 20 orbits and a total area of approximately 28 000 km^2. Constructing those two kinds of image blocks is to test: 1. if the elevation constraint works under different intersection conditions, 2. if the constraint will reveal the remarkable accuracy difference between the normal image and the anomalous one, when the images are well intersected. Table I provides a detailed description of the data used in this study.

The study area is located in north-eastern Shandong Province, China. The terrain is relatively flat and is dominated by plain terraces, with elevations varying from −5 to 30 m. Fig. 8 presents the distribution of images across the study area.

Based on the image data, we simulated two types of observation anomalies within stereo and planimetric image blocks over the study area to assess the effectiveness and robustness of our method under different intersection conditions. Next, we examined the effects of different compensation models–the translation, affine transformation, and quadratic polynomial models–on the detection rate. Building on the two simulated anomaly scenarios, we introduced multiple reference images in both image blocks and compared the anomaly detection performance between the multiple-reference and single-reference configurations. Finally, we simulated three additional types of observation anomalies: a single weakly connected image,

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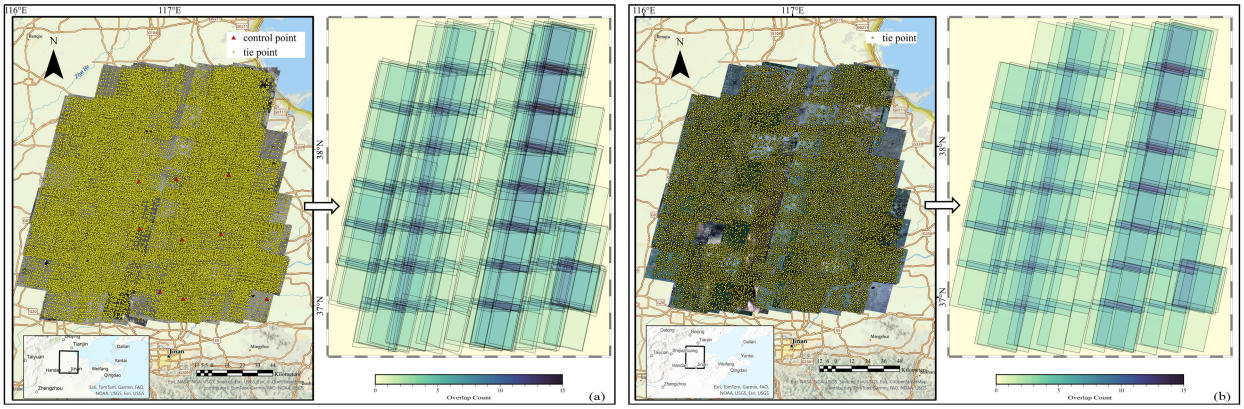


Fig. 8. Research area and the overlap of images. (a) Stereo image block and its overlap. (b) Planimetric image block and its overlap.

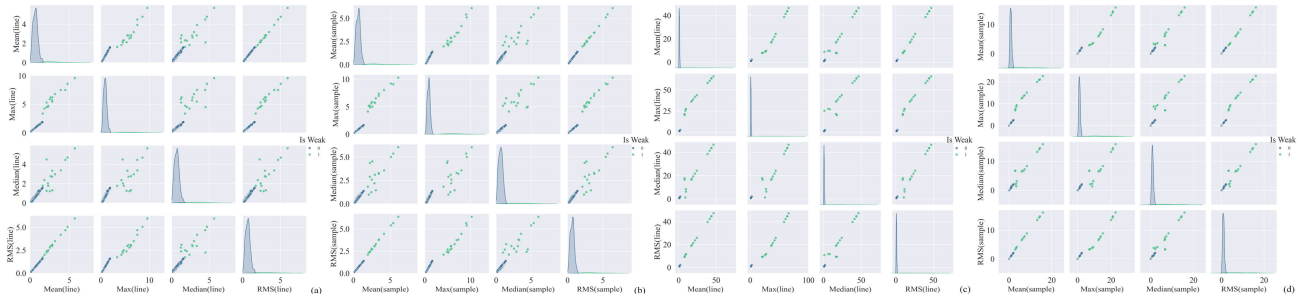


Fig. 9. Schematic of DBSCAN clustering results in two-feature spaces (blue dots: normal images; green dots: anomalous images) (unit: pixel). (a) and (b) Detection results of the stereo image block using the line and sample direction statistics. (c) and (d) Detection results of the planimetric image block using the line and sample direction statistics.

multiple weakly connected images, and multiple disconnected images. Controlled and uncontrolled block adjustments were conducted on both image blocks, with and without these anomalies. The resulting positioning accuracies were analyzed to quantify the influence of observation anomalies on block adjustment. Based on this evaluation, the anomalous images were further optimized through the addition of tie points.

B. Optimal Feature Space Determination

To select appropriate features for the DBSCAN algorithm, we calculated the mean, median, maximum, and RMSE of the geometric accuracy in both the line and sample directions for each image in the planimetric and stereo image blocks. Based on these statistics, various 2-D feature spaces were constructed by pairing the metrics, and DBSCAN was applied to cluster the images, with each image treated as a sample point. The detection rates of anomalous images were then evaluated. To further enhance the ability of DBSCAN to distinguish anomalous and normal images, we constructed feature spaces specifically for a case in which multiple weakly connected images are found in both image blocks, and conducted observation anomaly detection.

Fig. 9 presents the results: image samples exhibited significantly different distributions across feature spaces composed of different pairs of geometric accuracy statistics, leading to varying DBSCAN clustering results. For the planimetric image block, DBSCAN consistently and effectively clustered anomalous images across all feature spaces. In the stereo image block, DBSCAN performed well in the mean–maximum and median–maximum feature spaces, successfully separating anomalous and normal images. However, in the mean–median and median–RMSE feature spaces, separation was not satisfying, in spite of the correct identification of >90% of anomalous images. In contrast, the mean–RMSE feature space was the least effective, with 68% of anomalous images remaining undetected.

To ensure that the selected feature space would capture as much statistical information about geometric accuracy as possible, the mean–maximum feature space was ultimately adopted. The mean value represents the expected geometric accuracy across grid points, while the maximum value reflects the dispersion, thus jointly characterizing the geometric accuracy of each image.

C. Detection of Different Observation Anomalies

We designed two types of observation anomalies within planimetric and stereo image blocks in the study area and constructed three categories of image blocks: one without any anomalies, one containing multiple disconnected images plus a weakly connected image, and one containing multiple weakly connected images. In each block, we selected the central image as the reference image to prevent misdetection of anomalies caused by excessive geometric accuracy degradation due to the image position. Anomalous images and reference images were added to the planimetric and stereo image blocks, as shown in Fig. 10.

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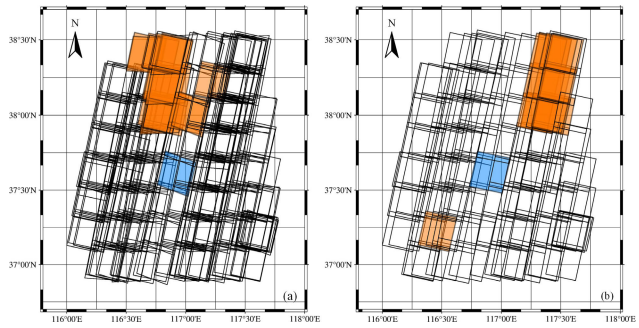


Fig. 10. Schematic of the spatial distribution of anomalous and reference images added to the image blocks (orange rectangles: multiple weakly connected images/disconnected images, transparent rectangles: normal images, blue rectangles: reference images). (a) Stereo image block. (b) Planimetric image block.

The results show that our method works even when the intersection angle reaches around 0°. Fig. 11 shows that there are obvious theoretical accuracy differences between the stereo image block and the planimetric image block; this demonstrates the effectiveness of the proposed method. The results also revealed substantial discrepancies in geometric accuracy between anomalous and normal images (Fig. 11). Disconnected images exhibited extremely poor geometric accuracy, on the order of several thousand times worse than normal, because they lacked the tie points with other normal images necessary to propagate the geometric accuracy of the reference image. Conversely, weakly connected images maintained significantly higher accuracy, with increases of approximately 140-fold in the line direction and 460-fold (for the stereo image block) and 670-fold (for the planimetric image block) in the sample direction. These findings demonstrate that even sparse tie points enable effective propagation of the geometric accuracy of the reference image. For the stereo image block, the geometric accuracy of weakly connected images was higher in the line direction than in the sample direction, due to the high degree of overlap between images in the line direction. This phenomenon allowed greater tie point matching, which led to greater propagation of reference image geometric accuracy in that direction. Overlap in the cross-track direction is typically low, and the acquisition time difference for the images is usually longer than 10 h, such that the number of tie points remains relatively small, and it is difficult for the geometric accuracy of the reference image to propagate well into other images in this direction. Notably, the geometric accuracy of the stereo image block was comparable to that of the planimetric image block in the sample direction, but higher in the line direction. This result was caused by the intersection angle of the forward- and backward-looking cameras, which was about 30°, such that the elevation constraint played a large role. In contrast, images in the planimetric image block were nearly nadir-facing, with an intersection angle close to 0°. Therefore, the elevation constraint is difficult to enforce effectively, leading to a severe decrease in the accuracy.

In all, our method accurately identified all anomalous images, even when multiple types coexisted with markedly different geometric accuracies; its effectiveness is validated.

D. Detection Rate Under Different Compensation Models

To further investigate how the correlation between model parameters and pixel positions affects observation anomaly detection, we applied three compensation models (translation, affine transformation, and quadratic polynomial) to the image blocks defined in Section V-C, and performed observation anomaly detection based on the corresponding image blocks. We quantitatively compared the detection performance of the three models across two types of image blocks using confusion matrices, as shown in Tables II and III.

The results revealed that some normal images were misclassified as anomalous and vice versa. We quantified this behavior using the true positive rate as $TPR = TP/(TP+FN)$, where TP and FN are the numbers of true positives (actual anomalous images correctly identified) and false negatives (anomalous images incorrectly flagged as normal), respectively, the false alarm rate as $FAR = FP/(FP + TN)$, where FP and TN are the numbers of false positives (normal images incorrectly flagged as anomalous) and true negatives (normal images correctly identified), respectively, and the false negative rate as $FNR = FN/(FN + TP)$. Consequently, TPR reflects the proportion of anomalous images correctly detected, FAR represents the proportion of normal images misclassified as anomalous, and FNR indicates the proportion of anomalous images misclassified as normal (Tables IV and V).

When the translation model was applied, alomst all anomalous images in both image blocks were detected, achieving accuracy comparable to that of the affine transformation model. However, because the translation model has only two parameters and is independent of the pixel position, its parameter accuracy is actually the model accuracy, which depends solely on the image block configuration. Consequently, it may struggle to detect anomalous images when the tie point distribution is uneven.

Even when weakly connected images in the planimetric image block were detected, their geometric accuracy differed only marginally from that of normal images. The average geometric accuracy of normal images was 0.06 pixels in both the line and sample directions, whereas that for weakly connected images averaged 0.58 pixels in the line direction and 0.62 pixels in the sample direction; all disconnected images averaged 291.98 and 4.32 pixels, respectively, which were approximately 480- and sevenfold higher than those of weakly connected images, and 4800-fold and 70-fold higher than those of normal images. The small difference between weakly connected images and normal images increased the risk of misclassification, making the translation model unsuitable for detecting observation anomalies.

With the quadratic polynomial model, detection accuracy for one weakly connected and multiple disconnected images was comparable to that of the affine transformation model; the quadratic polynomial model successfully identified both observation anomaly types in both stereo and planimetric image blocks. However, it performed poorly in scenarios with multiple weakly connected images; in the stereo image block, 6 anomalous images were missed (false negative rate; FNR = 38%), and 17 normal images were falsely flagged

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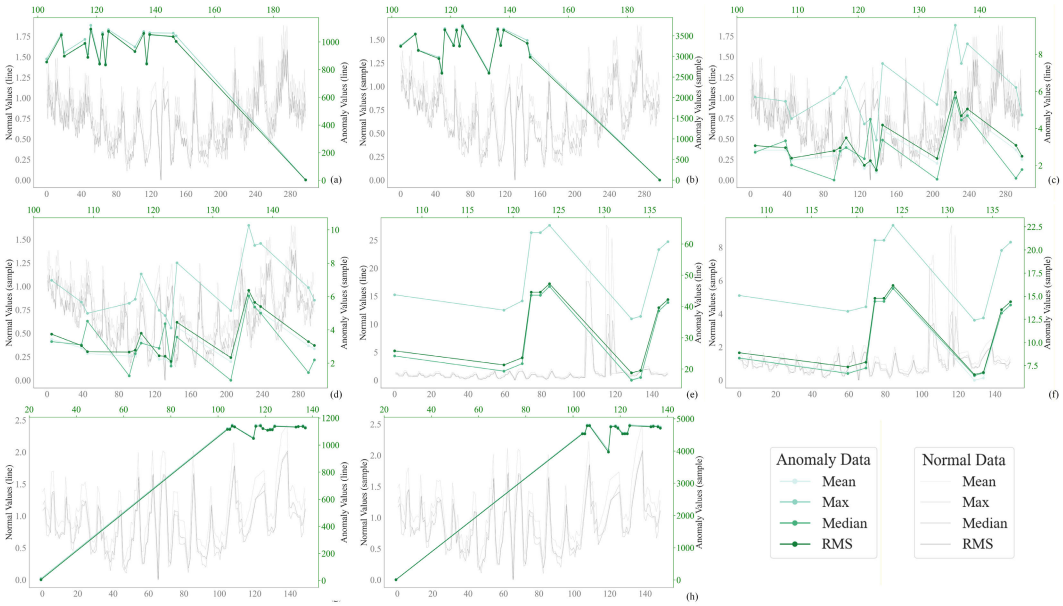


Fig. 11. Geometric accuracies (the line and sample directions) of anomalous and normal images in stereo and planimetric image blocks as the affine transformation model is applied (gray lines: anomalous images geometric accuracies, green and blue lines: images' images geometric accuracies) (unit: pixel). (a) and (b) Stereo image block with multiple weakly connected images. (c) and (d) Stereo image block with multiple disconnected images. (e) and (f) Planimetric image block with multiple weakly connected images. (g) and (h) Planimetric image block with multiple disconnected images.

TABLE II

DETECTION RESULTS IN THE STEREO IMAGE BLOCK USING THREE IMAGE-SPACE COMPENSATION MODELS. NORMAL AND ANOMALOUS IN ROW: THE TRUTH VALUE, *NORMAL* AND *ANOMALOUS* IN COLUMN: THE PREDICTION

		Translation		**Affine**		**Quadratic**	
		Anomalous	Normal	Anomalous	Normal	Anomalous	Normal
One weakly connected image and multiple disconnected images	Anomalous	16	1	17	0	17	0
	Normal	0	281	0	281	0	281
Multiple weakly connected images	Anomalous	16	0	16	0	10	6
	Normal	0	282	0	282	17	265

TABLE III

DETECTION RESULTS IN THE PLANIMETRIC IMAGE BLOCK USING THREE IMAGE-SPACE COMPENSATION MODELS. NORMAL AND ANOMALOUS IN ROW: THE TRUTH VALUE, *NORMAL* AND *ANOMALOUS* IN COLUMN: THE PREDICTION

		Translation		**Affine**		**Quadratic**	
		Anomalous	Normal	Anomalous	Normal	Anomalous	Normal
One weakly connected image and multiple disconnected images	Anomalous	17	0	17	0	17	0
	Normal	0	132	0	132	0	132
Multiple weakly connected images	Anomalous	16	0	16	0	10	6
	Normal	0	133	0	133	6	127

TABLE IV

TPR, FA, AND FNR OF OBSERVATION ANOMALY DETECTION RESULTS OF THE STEREO IMAGE BLOCK, THREE IMAGE-SPACE COMPENSATION MODELS ARE APPLIED

	Translation			**Affine**			**Quadratic**		
	TPR	FA	FNR	TPR	FA	FNR	TPR	FA	FNR
One weakly connected image and multiple disconnected images	0.94	**0.00**	0.06	**1.00**	**0.00**	**0.00**	**1.00**	**0.00**	**0.00**
Multiple weakly connected images	**1.00**	**0.00**	**0.00**	**1.00**	**0.00**	**0.00**	0.62	0.06	0.38

(FAR = 6%); in the planimetric image block, the FNR was 38%, and the FAR was 4%.

As shown in Fig. 12, the geometric accuracy metrics for both normal and anomalous images fluctuated

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TABLE V

TPR, FA, AND FNR OF OBSERVATION ANOMALY DETECTION RESULTS OF THE PLANIMETRIC IMAGE BLOCK, THREE IMAGE-SPACE COMPENSATION MODELS ARE APPLIED

	Translation			Affine			Quadratic		
	TPR	FA	FNR	TPR	FA	FNR	TPR	FA	FNR
One weakly connected image and multiple disconnected images	**1.00**	**0.00**	**0.00**	**1.00**	**0.00**	**0.00**	**1.00**	**0.00**	**0.00**
Multiple weakly connected images	**1.00**	**0.00**	**0.00**	**1.00**	**0.00**	**0.00**	0.62	0.04	0.38

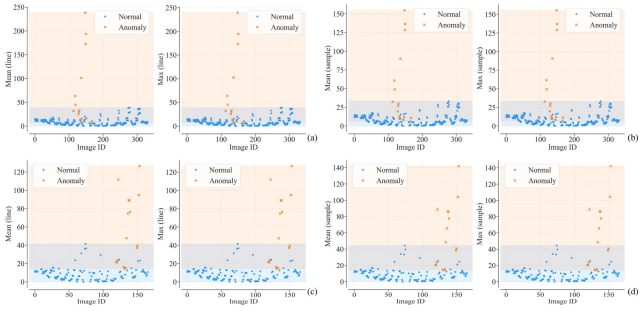


Fig. 12. Observation anomaly detection results of the image blocks using the quadratic polynomial model (blue dots: normal images, orange crossings: anomalous images) (unit: pixel). (a) and (b) Mean and maximum of geometric accuracies of all stereo images (line and sample directions). (c) and (d) Mean and maximum of geometric accuracies of all planimetric images (line and sample directions).

significantly under the quadratic model, complicating the observation anomaly detection. We attribute this result to over-parameterization; the excessive degrees of freedom of the quadratic polynomial model introduced instability in the parameter estimation, causing the geometric accuracy of both image types to overlap and reducing the discriminative power of the model.

E. Influence of the Reference Image Number on Detection Results

In the experiments described in Section V-C, anomalies were detected by setting the central image as the sole reference, with the expectation that anomalous images would exhibit significantly lower geometric accuracy. However, some normal images also exhibited relatively low geometric accuracy due to their distance from the reference image. Thus, while single-reference detection remains feasible, it may be ineffective, particularly for large image blocks populated with multiple weakly connected images. One promising solution to this problem is to deploy multiple reference images distributed across the block. Analogous to uniformly placed GCPs in block adjustment, uniformly distributed reference images offer robust geometric control, ensuring the accurate differentiation of normal images and anomalous images.

To validate this method, we reconfigured the stereo and planimetric image blocks to include multiple weakly connected images and evaluated the detection performance of the proposed method using multiple reference images. Detection failed with only the central reference in the stereo block, as some weakly connected anomalies were overlooked, and some normal images were falsely flagged. Next, we added references at the four corners of the image block, alongside the center (Fig. 13). Adding references in the bottom corners did not improve detection performance due to decreased accuracy propagated from these references. However, adding references in the top corners nearly eliminated false detections, except for a single misclassification at the bottom left corner, attributable to a loss of geometric accuracy during propagation.

Similarly, in the planimetric block, the central reference image detected only a subset of anomalous images and incorrectly flagged some normal images. Adding references in the top corners and the bottom right corner failed to fully resolve the issue, as the accuracy propagated from these references became comparable to that of weakly connected images. Adding a bottom-left reference eliminated misclassifications but still missed some anomalous images. A satisfactory result was achieved only when references were simultaneously placed in all four corners and the center, reducing misclassification to only two normal images.

These experiments demonstrated that in scenarios involving multiple weakly connected anomalies, a sufficiently large and uniformly distributed set of reference images is essential for maximizing detection performance.

F. Effects of Observation Anomalies on Block Adjustment Results

We investigated the effects of observation anomalies on block adjustment by comparing two types of image blocks derived from the study area, based on the stereo image block: an entirely uncontrolled image block (without GCPs) and a controlled image block with nine evenly distributed GCPs. We introduced three types of observation anomalies into the uncontrolled block: one weakly connected image, multiple weakly connected images, and multiple disconnected images. Using the GCP-controlled image block as the reference, we calculated the positioning accuracy of tie points for each image across the three uncontrolled blocks, computing both mean and RMSE values in the line and sample directions to assess deviation from the reference image block. The reason for not directly using GCPs to evaluate the impact is: first, the number and distribution of the intersected tie points meet the requirement; there are hundreds or thousands of them almost evenly distributed in the image blocks, providing adequate statistical information and ensuring the reliability of the assessment. Second, the intersected tie points are obtained by automatic image matching; none or only a few human forces are needed to conduct the assessment (e.g.,

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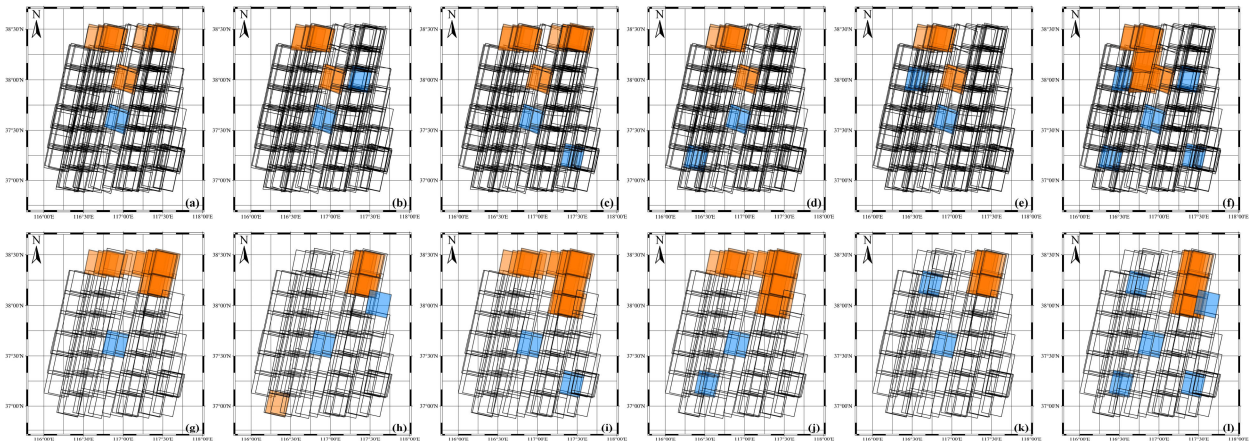


Fig. 13. Observation anomaly detection results after increasing the reference image numbers, in the stereo and planimetric image blocks (blue rectangles: reference images, orange rectangles: anomalous images, transparent rectangles: normal images). (a)–(f) Detection results of the stereo image block. (g)–(l) Detection results of the planimetric image block.

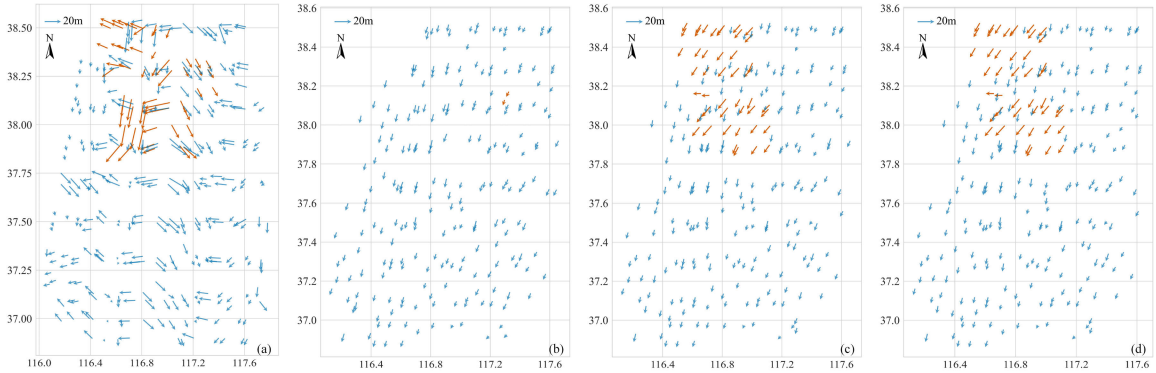


Fig. 14. Impact of observation anomalies on block adjustment (blue arrows: positioning errors of normal images, red arrows: positioning errors of anomalous images) (unit: m). (a) Before block adjustment. (b) After uncontrolled block adjustment, a weakly connected image exists. (c) After uncontrolled block adjustment, multiple weakly connected images exist. (d) After uncontrolled block adjustment, multiple disconnected images exist.

TABLE VI

POSITIONING ERRORS OF THE THREE TYPES OF IMAGE BLOCKS BEFORE AND AFTER UNCONTROLLED BLOCK ADJUSTMENT (UNIT: M)

		Mean value				**RMSE**			
		Before		After		Before		After	
		Line	Sample	Line	Sample	Line	Sample	Line	Sample
With one weakly connected image	Anomalous images	-13.62	5.28	-7.48	-2.63	-	-	-	-
	Normal images	-10.19	-2.63	-8.80	-2.46	11.23	8.52	1.60	1.58
	All images	-10.21	-2.61	-8.80	-2.46	11.21	8.51	1.60	1.58
With multiple weakly connected images	Anomalous images	-15.85	-7.90	-12.99	-7.29	9.13	10.22	0.79	1.08
	Normal images	-9.89	-2.31	-8.58	-2.09	11.23	8.31	1.51	1.66
	All images	-10.21	-2.61	-8.81	-2.37	11.21	8.51	1.78	2.01
With multiple disconnected images	Anomalous images	-15.85	-7.90	-13.23	-7.51	9.13	10.22	0.89	0.88
	Normal images	-9.89	-2.31	-8.46	-2.11	11.23	8.31	1.53	1.69
	All images	-10.21	-2.61	-8.71	-2.40	11.21	8.51	1.85	2.05

eliminating possible outliers). It is time-saving, cost-saving, and makes it possible to yield reliable assessment results. This approach allowed us to evaluate the influence of observation anomalies on absolute positioning accuracy in uncontrolled block adjustments. Fig. 14 presents the three types of image blocks with corresponding observation anomalies.

The results from Table VI revealed that uncontrolled block adjustment consistently improved geometric accuracy across all images in the three scenarios. The largest mean improvement reached 1.50 m in the line direction and 0.24 m in the sample direction; RMSE improved by up to 9.61 m in the line direction and 6.93 m in the sample direction. These modest shifts in mean value reflect the differing convergence targets; the GCP-controlled block adjustment aligned to external references, whereas uncontrolled image blocks aligned based solely on tie points, causing systematic offsets. The more substantial RMSE reductions demonstrate improved intersection conditions after the block adjustment. Notably,

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TABLE VII
POSITIONING ERROR STATISTICS COMPARISON BEFORE AND AFTER THE OPTIMIZATION (UNIT: M)

	Mean value				RMSE			
	Before		After		Before		After	
	Line	Sample	Line	Sample	Line	Sample	Line	Sample
Anomalous images	-12.99	-7.29	-10.27	-1.51	0.79	1.08	0.79	0.43
Normal images	-8.58	-2.09	-8.66	-2.27	1.51	1.66	1.52	1.60
All images	-8.81	-2.37	-8.75	-2.23	1.78	2.01	1.54	1.57

the line-direction positioning error remained larger than that in the sample direction, suggesting that uncontrolled block adjustment effectively mitigated cross-epipolar errors but left along-epipolar errors due to the lack of external control.

For the image block containing one weakly connected image, the mean positioning error of the anomalous image decreased significantly after block adjustment, from −13.62 to −7.48 m in the line direction and from 5.28 to −2.63 m in the sample direction (Table VI). These metrics are comparable to those of normal images, indicating that even a few unevenly distributed tie points can sufficiently leverage spatial connection to enhance positioning accuracy.

With regard to multiple weakly connected images, improvements were observed in both anomalous and normal images. Comparison of the mean and RMSE of positioning errors between anomalous and normal images revealed substantial discrepancies, with mean error differences of 4.41 m in the line direction and 5.20 m in the sample direction, while the corresponding RMSE differences reached −0.72 and −0.58 m, respectively. These results were attributed to the weak spatial connection of anomalous images, which relies heavily on internal tie points, causing their block adjustment results to converge toward different references compared to other images. Consequently, their mean and RMSE values diverge. The image block with multiple disconnected images yielded similar results and will not be discussed further.

G. Positioning Precision Improvement After Observation Condition Optimization

As demonstrated in Section V-F, the presence of observation anomalies within an image block can degrade local positioning accuracy and contaminate the overall block adjustment result, regardless of whether block adjustment is performed with or without ground control. Therefore, it is necessary to improve the relative positioning accuracy of anomalous images and maintain consistency in the overall accuracy of the image block. A practical and effective approach is to increase the number of tie points in anomalous regions. It strengthens spatial connections within the image block and improves the consistency of positioning accuracy.

In the present study, we added approximately 1300 tie points to the moderately weakly connected images in the stereo image block. Because tie points represent the spatial constraint between overlapping images, a dense and uniform tie point configuration is expected to improve the alignment of anomalous images with the rest of the block. To assess its effectiveness, we evaluated the adjustment results under an uncontrolled block adjustment scenario, as shown in Fig. 15.

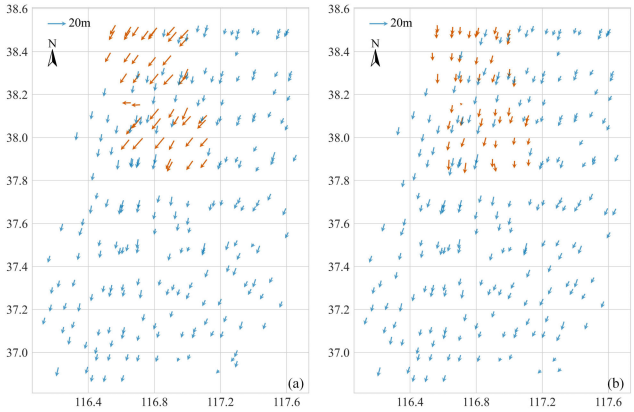


Fig. 15. Positioning error improvement after the optimization (blue arrows: positioning errors of normal images, red arrows: positioning errors of anomalous images) (unit: m). (a) Before adding tie-points. (b) After adding tie-points.

Our results revealed marked improvements in the relative positioning accuracy of the anomalous images (Table VII). The introduction of additional tie points led to significant reductions in the positioning errors of these images, with mean positioning errors reduced by 2.72 m in the line direction and 5.78 m in the sample direction, respectively. These values approach the positioning accuracy levels of normal images, suggesting that the spatial relationships between anomalous and normal images were effectively improved. These findings confirm that the incorporation of additional tie points is a viable strategy for mitigating the negative effects of observation anomalies and achieving a consistent block adjustment result.

V. DISCUSSION

A. Time Cost of Different Compensation Models

From our previous experiments, the more coefficients of the compensation model, the more time is needed to detect the anomalous images. It is because the number of the image-space compensation model coefficients directly affects the size of the coefficient matrices of the intersection constraint, namely the $\boldsymbol{A}_{tp}$ in (9).

If the image-space compensation model is a translation model, and there are N_{img} images in the image block, the size of $\boldsymbol{A}_{tp}$ will be $N_{img} \times 2N_{img} = 2N_{img}^2$. Similarly, if using the affine transformation model, the size of $\boldsymbol{A}_{tp}$ will be $6N_{img}^2$ and if using the quadratic polynomial model, it will be $12N_{img}^2$. The detailed computation time cost of these compensation models is listed in Table VIII.

Although the ratios of the parameter numbers are translation: affine: quadratic = 1:3:6, while the execution time

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TABLE VIII
TIME COST OF DIFFERENT COMPENSATION MODELS

	Stereo block	Planimetric block
Translation	136.93 s	254.17 s
Affine	4642.96 s	438.39 s
Quadratic	66072.94 s (~18.35h)	4789.76 s (~1.3h)

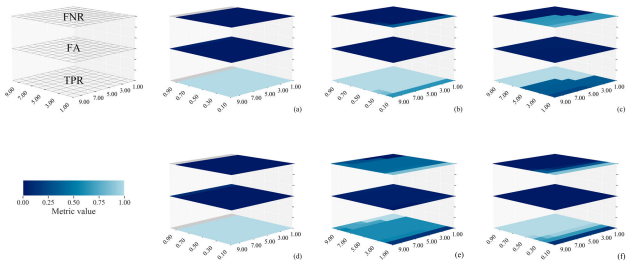


Fig. 16. Heat maps of detection metrics under different ε – min _*sample* combinations of the stereo [see (a)–(c)] and planimetric [see (d)–(f)] image blocks. (a) and (d) Using a translation model. (b) and (e) Using an affine transformation model. (c) and (f) Using quadratic polynomial model (gray areas mean NaN values).

ratio is different. Similarly, the ratio of images numbers of these two image blocks is 2:1, but the execution time ratios are also different under the same image-space compensation model. The above results also explain why we do not recommend using the quadratic polynomial model as the image-space compensation model, due to its great computational cost.

B. Practical Method to Improve Positioning Accuracy of Anomalous Areas

In our current work, the positioning accuracy optimization of anomalous areas is achieved by simulation. In practice, it is feasible to use both automatic image matching and manual matching methods to add tie-points to anomalous areas, thus optimizing the positioning accuracy of those areas. If it is the ground truth causing the observation anomaly (e.g., the shore area), the targeted image matching methods may obtain more reliable tie-points than the general algorithms or pipelines (e.g., coastline/shoreline matching, frequency-domain matching [33], deep-learning models [34], and so on). If it is the time phase difference causing the observation anomaly, it will be better to manually mark tie-points on the anomalous areas to reconstruct image connections. If neither of these two methods fails to optimize the connection relationship between anomalous and normal images, it may be possible to use high-level image products to yield high-accuracy positioning results.

C. Influence of Hyperparameters on the Detection Result

To find out the influence of hyperparameters on the detection result, we test different ε – min _*sample* combinations on the image block with the above-mentioned observation anomaly, and the results are as Fig. 16.

It is clear that for the translation model and the affine transformation model, we can always find a ε – min _*sample* combination to yield the most satisfying detection results (TPR = 1, FA = 0 and FNR = 0). Thus, for the translation model, it is ε = 0.2, min _*sample* = 5 for both the stereo and the planimetric image block. For the affine transformation model, it is ε = 0.2, min _*sample* = 5 for both the stereo and the planimetric image block. For the quadratic polynomial model, there are actually no perfect combinations to yield satisfying results, we can only find a relative optimal combination: ε = 5, min _*sample* = 4 for the stereo image block and ε = 8, min _*sample* = 3 for the planimetric image block.

VI. CONCLUSION

We proposed a method for detecting observation anomalies in image blocks based on the uncertainty propagation law and clustering analysis. The method involves establishing a reference image geometric accuracy model and constructing intersection and elevation constraints. By calculating the geometric accuracy of the image-space compensation model for each image using the uncertainty propagation law, we computed statistical metrics (mean, median, maximum, and RMSE) for each image within an image block. Paired analysis of these metrics identified the mean–maximum combination as the optimal. A corresponding feature space was modeled on this combination, and the DBSCAN algorithm was applied to detect anomalous images. Three image-space compensation models were applied to detect anomalous images, and the affine transformation model yielded satisfactory results (TPR = 100%, FAR = 0%, FNR = 0% for all observation anomaly types). We determined that in a large-scale image block with relatively moderate observation anomalies, one reference image may be insufficient due to geometric accuracy decay; therefore, up to four reference images were added to different corners of the image blocks. The results confirmed the feasibility of the method.

To further explore the effects of observation anomalies on the image block, we calculated the positioning errors of uncontrolled block adjustment of images blocks for different types of observation anomalies. The results revealed that observation anomalies significantly influenced positioning accuracy. To address this issue and ensure consistent positioning accuracy for the image block, we introduced additional tie points to the anomalous regions and obtained satisfactory results. In summary, we developed a novel observation anomaly detection method that leverages the uncertainty propagation law and the DBSCAN clustering algorithm, creating a quality-control framework for satellite image block adjustment that does not rely on GCPs. The method is time-saving and cost-effective, and fills the current gap in quality control methods for the automatic and accurate detection of observational anomalies.

In future work, we will detect the exact locations of anomalous regions within the image block. In our current work, the detection target is the image. When detecting the anomaly areas, we plan to generate evenly distributed virtual image points based on the whole image block, the same as what we did for the reference image. This time, the density of the virtual

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points is a key parameter, since it determines how possible and precise the anomalous areas can be found. Also, the feature space will possibly be reconstructed, and the clustering method might be changed since the sample's distribution will change.

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